Evanns Morales-Cuadrado

Robotics PhD Candidate | U.S. Citizen | Fulbright Specialist
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SUMMARY

Robotics PhD candidate in Dr. Samuel Coogan's FACTS Lab at Georgia Tech, Fulbright Specialist, and year-round AI for Autonomy intern at Sandia National Laboratories. Specializes in safe autonomy, reinforcement learning, and real-time control. Proven track record of developing and deploying safe, computationally efficient planning and control methods for autonomous systems, with work published in leading peer-reviewed robotics and controls venues. Selected to the U.S. Department of State's Fulbright Specialist Roster after spearheading an international research collaboration with a Brazilian partner institution and securing research funding for the work.

EDUCATION

Georgia Institute of Technology 2022-Present
Ph.D. in Robotics; Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington 2018-2022
Honors Bachelor of Science in Electrical Engineering; Minor in Mathematics; Minor in Physics; Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Sandia National Laboratories, Albuquerque, NM May 2026 - Present
Year-Round Graduate Research Intern, AI for Autonomy

- Developed novel reinforcement learning-based quadrotor-vs-quadrotor interception strategies, reaching targets 2.4x sooner with 14.7x more impact energy than baselines
- Created a novel Newton-Raphson (NR)-based interception law, raising interception success from 10% to 70% over standard NR
- Introduced novel benchmarking for the terminal characteristics of quadrotor interceptions
- Built C#/.NET networking frameworks and execution scripts that coordinate multi-robot communication and task execution

Georgia Institute of Technology, Atlanta, GA August 2022 - Present
Graduate Research Assistant, FACTS Lab (Advisor: Dr. Samuel Coogan)

- Deployed a novel Newton-Raphson-based tracking control law on a quadrotor, cutting error 59% on average versus PX4's native tracker
- Outperformed NMPC on a quadrotor (5x faster computation, 3.6x less CPU energy) and a blimp (65x faster computation, 16x less CPU energy), with competitive quadrotor accuracy and 2.2x lower blimp error
- Created RTD-RAX, a receding-horizon safe trajectory planner that plans less conservatively than baselines and stays safe in high-disturbance scenarios where baselines crash
- Optimized and deployed Gaussian-process runtime assurance on a quadrotor for a 100 Hz control loop, landing safely in unknown wind from an industrial fan
- Built the lab's real-time flight-test stack, enabling repeatable hardware validation

Georgia Institute of Technology, Atlanta, GA January 2024 - May 2026
Teaching Assistant, Vertically Integrated Project in Robotics

- Mentored 20 undergraduates per semester in hands-on robotics research, several of whom went on to robotics PhD programs
- Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on robotics and autonomous control systems

University of Texas at Arlington, Arlington, TX November 2021 - May 2022
Research Assistant, Autonomous Systems Lab (Advisor: Dr. Frank Lewis)

- Trained in one of the most-cited control research groups in electrical engineering, with Dr. Lewis' work ranked 12th in the USA and 23rd worldwide by citation impact
- Contributed to reinforcement learning-based quadrotor control research

University of Texas at Arlington, Arlington, TX January 2022 - May 2022
Teaching Assistant, Graduate-Level Intelligent Systems Course

- Selected as an undergraduate teaching assistant for a graduate-level course
- Led exam review sessions and graded homework, exams, and projects

University of Texas at Arlington, Arlington, TX October 2019 - May 2022
Research Assistant, Dynamical Networks and Control Lab (Advisor: Dr. Yan Wan)

- Completed UTA Honors College thesis on learning-based minimum-time and minimum-energy multi-vehicle path planning

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

SELECTED PUBLICATIONS

E. Morales-Cuadrado, S. Coogan, Z. Kakish, and A. Parikh. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA). (title redacted until the double-blind review process is complete)*

L. Pinto*, **E. Morales-Cuadrado***, S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA). *Equal contribution. (title redacted until the double-blind review process is complete)*

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, "Trajectory Tracking for Systems with Unknown Time-Varying Disturbances." *IEEE Open Journal of Control Systems (OJ-CSYS)*, 2026.

E. Morales-Cuadrado, L. Chung, S. Kousik, and S. Coogan, "`RTD-RAX`: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances." Joint Submission to 2026 MECC-ALDSC. *(accepted to both) preprint available.*

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, "Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic." Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. *(under revision) preprint available.*

E. Morales-Cuadrado, L. Baird, Y. Wardi, and S. Coogan, "Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method." *IEEE Transactions on Control Systems Technology (TCST)*, 2026. *preprint available.*

E. Morales-Cuadrado, C. Llanes, Y. Wardi, and S. Coogan, "Newton-Raphson Flow for Aggressive Quadrotor Tracking Control." *2024 American Control Conference (ACC). preprint available.*

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech's Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

Georgia Institute of Technology, Atlanta, GA — August 2024 - Present
Co-Founder and President of Puerto Rican Student Association

- Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture
- Formed a three-member co-founder board and identified a faculty advisor
- Raised funds for the organization and recruited members from campus
- Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech

HONORS AND AWARDS

Fulbright Specialist — August 2026
Peer-review-selected to the U.S. Department of State's Fulbright Specialist Roster for a three-year term

Top-3 Finalist at a Deep Learning Research Symposium — November 2023
Professor-sponsored award for novel research in deep learning and robotics

Goizueta Fellowship at Georgia Tech — August 2022
Highly selective fellowship for graduate students of Latin-American descent

Summa Cum Laude Honors at University of Texas at Arlington — May 2022
Exceptional academic distinction in top GPA tier for graduation honors

Electrical Engineering Honors Scholar at University of Texas at Arlington — December 2021
Inaugural member of the electrical engineering honors cohort

IEEE-HKN International Electrical Engineering Honor Society Membership — August 2021
Selective undergraduate society recognizing outstanding academic achievement and community service

Chance Vought Engineering and Science Endowment Scholarship — August 2020
Highly selective award given annually to three recipients across the entire College of Engineering

National Merit Scholar — March 2018
Recognition of top national performers on the SAT